Lab 1 – Grad Pace Product Description

GradPace – Team Copper

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1 Introduction

- Workforce expectations are rapidly evolving
- 63% of employers cite skill gaps as top barrier to growth [World Economic Forum, 2025]
- 66% of enterprises are reducing entry-level hiring due to AI [Laurent, 2026]
- 91% report automation is changing or eliminating jobs [Laurent, 2026]
- 50-60% of recent graduates are underemployed [Cengage Group, 2025]
- Problem exists in higher education, career services, and entry-level job markets
- 48% of recent graduates feel underqualified for entry-level roles [Cengage Group, 2025]
- [Figure 1: Fluctuations in perception of preparedness in recent graduates]
- Academic progress is tracked, but career preparation is fragmented
- Current tools address isolated needs only
    - Coursework tracking (students can track their academic data, but not professional information)
    - Job boards
    - Networking platforms
- Gap: no unified, real-time career readiness roadmap
- Key problem characteristics
    - Fragmented data career development systems
    - Delayed gap detection
    - Reactive career decision-making
    - Limited visibility into career readiness

- [Figure 2: Educator POV on teaching responsibilities]
- Introduce GradPace as a unified, career readiness platform

2 Product Description

GradPace is a career readiness platform built to close the gap between student's academic progress and their transition to professional employment. Built by Team Copper for students, alumni, and industry professionals, the platform is designed around proactive preparation rather than the reactive, one-size-fits-all approach of traditional job boards.

Students start by filling out their academic and professional information. GradPace generates a personalized roadmap that lines up their current progress with recommendations based on real industry standards. This works through two core capabilities, early-warning gap detection and an industry readiness dashboard, both built to catch what a student needs long before they hit the job market. The platform pulls in live industry data and includes GPS-based local networking to keep that guidance grounded and relevant.

As a result, users are not just reflecting on past work they have done but staying aligned with what the modern workforce expects from new grads. ß

2.1 Key Product Features and Capabilities

GradPace brings together a full set of features that carry a student from their first login all the way through active career preparation. It starts with account management, covering login authentication, account creation, and subscription controls, so every user has secure, straightforward access to the platform. From there, students can input their academic and professional background through a dedicated interface built specifically to capture the details that matter for their goals, coursework, projects, experience, and anything else that feeds into their profile.

Once that information is in the system, GradPace's personalized roadmap generation takes over, building a custom plan for each student based on their actual data rather than generic advice. Students can track that plan through the view roadmap dashboard, a visual, easy-to-follow interface that shows progress over time. What really sets the platform apart, though, is early-warning gap detection, one of GradPace's most distinctive features, which catches missing skills or experience in a student's profile early enough that they can do something about it. Paired with the industry readiness dashboard, which measures a student's standing against real-world job requirements, students get a clear, ongoing sense of how close they are to being job-ready, instead of finding out after the fact.

Beyond the core roadmap tools, GradPace also supports the broader career development experience. Location-based event recommendations surface relevant networking events and opportunities near the student based on their physical location, helping turn local connections into real career leads. Alumni play a role too, through the alumni survey, which gathers insight on current industry trends and in-demand skills directly from people who've already gone through the process, while also giving alumni a way to earn back part of what they spent on their subscription. All of this is backed by a steady stream of industry data pulled from real industry professionals, keeping the platform's recommendations current so students aren't building their plans around outdated advice.

2.2 Major Components (Hardware/Software)

GradPace operates as a web-based platform that requires almost any up-to-date hardware with access to internet to access cloud servers. Following the structural design of the CS 410 Major Functional Component Diagram (MFCD), the software architecture is divided into four core components: the presentation layer (frontend), application layer (backend), data layer (database), and API integrations. The frontend hosts the user interface for account management, information tracking, and dashboard navigation, while the backend processes system logic such as security, roadmap generation, gap detection, benchmark comparisons, and development recommendations. Database, including user profiles, academic and professional records, skills, and opportunity listings, is securely stored through a PostgreSQL database layer. Additionally, external API integrations enable the platform to connect with live industry data, location services, and AI systems for continuous roadmap refinement and real-time recommendations.

3 Identification of Case Study

- **WHO:** Alex, a computer science undergraduate student in his second semester of college
- **WHAT PROBLEM:** Unsure how to begin professional computer science development outside of general classwork.
- **FEATURES:** Opportunity finding, custom roadmap
- **SCENARIO:** Alex inputs transcripts, current work interests, and location information, and now has access to a step-by-step roadmap to help develop his skills to prepare for that post-grad goal

- **FUTURE:** University integration, alumni engagement, market trend data use to keep university courses relevant

Table 1: Table of Comparison Between RWP and Prototype

Feature	Prototype	Real World Product
Account Management	PC display	Live user database
Personal data input	From entry	Student profile
Roadmap	4 year plan depending on student goal and plan	AI driven personalized path
Gap Detection	Checklist	Realtime alerts with benchmarking
Local networking events	Simulated event list	GPS-based events

4 Glossary

- **API (Application Programming Interface):** A set of rules that allow different software applications to communicate and exchange data with each other.
- **Authentication**: The process of verifying a user's identity before allowing access to a system.
- **Backend**: The server-side part of a website or application that handles logic, databases, and data processing.
- **Career Readiness**: A measure of how prepared a student is to enter a specific career based on skills, experience, and qualifications.
- **Career Roadmap**: A structured, visual plan that outlines a student’s progress, goals, and required steps toward a specific career path.
- **Cloud Computing**: The use of remote servers on the internet to store data and run applications instead of a local computer.
- **Data Leak**: The accidental or unauthorized exposure of sensitive or private information.

- **Database**: An organized collection of data that can be stored, accessed, and managed electronically.
- **Encryption**: A security method that converts data into a coded form so that only authorized parties can read it.
- **Frontend**: The user-facing part of a website or application that users interact with in their browser.
- **Gap Detection (Early-Warning System):** A feature that identifies missing skills, experiences, or qualifications needed for a target career.
- **Industry Readiness Benchmarking**: The process of comparing a student's profile against expected skills and qualifications for specific careers.
- **Location-Based Event Recommendations**: Suggestions for nearby internships, networking events, or career opportunities based on user location.
- **Professional Data Input**: User-provided information such as coursework, projects, internships, and certifications used to build a profile.
- **Survey Results Dashboard**: A feature that displays aggregated insights from alumni data to inform students and administrators.
- **System Administrator**: A role responsible for managing the platform, maintaining data quality, and overseeing system functionality.
- **Token** (**AI**): A small unit of text processed by an AI model, such as a word, part of a word, or punctuation.
- **User Interface** (**UI**): The visual elements of a software application that allow users to interact with it.

- **Web Server**: A computer system or software that stores websites and delivers web page to users when they request them through a browser.

Reference

Brown, E. (2025, January 21). New Survey Reveals Traditional Undergraduate Education Is Not Preparing Students for the Workforce. Hult International Business School. https://www.hult.edu/blog/wi_skills_survey/

Cengage Group. (2025). The Missing Link: Accountability in Career Readiness. Cengage Group. https://www.cengagegroup.com/edtech-research/employability-report/

Laurent, A. (2026, March 8). AI's Impact on Graduate Jobs: A 2025 Data Analysis. IntuitionLabs. https://intuitionlabs.ai/articles/ai-impact-graduate-jobs-2025

Mardis, M. A., Ma, J., Jones, F. R., Ambavarapu, C. R., Kelleher, H. M., Spears, L. I., & McClure, C. R. (2018). Assessing Alignment Between Information Technology Educational Opportunities, Professional Requirements, and Industry Demands. Education and Information Technologies, 23(4), 1547–1584. https://doi.org/10.1007/s10639-017-9678-y

World Economic Forum. (2025, January 7). Future of Jobs Report 2025. World Economic Forum. https://www.weforum.org/reports/the-future-of-jobs-report-2025/